

Tarrun Pitta

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Education

California State University, Fullerton

Fullerton, CA

Master of Science in Computer Science

08/2024 - 05/2026

Publication: Hardware Trojan Detection with Machine Learning and Power Side-Channels: A Post-Deployment Analysis

Experience

Pacific Life

Newport Beach, CA

Software Engineer

06/2025 - 04/2026

- **Automated top-of-funnel underwriting risk evaluation** for **15,000+ daily loan records** by architecting and deploying an event-driven, serverless pipeline using **Python, AWS Lambda, and LangChain**.
- **Reduced policy research time by 96%**, from **15 minutes to 30 seconds per lookup**, by engineering a production-grade **Retrieval-Augmented Generation (RAG)** pipeline using **Pinecone** for high-performance semantic search.
- **Ensured 100% regulatory compliance for sensitive financial data (PII)** across cloud transactions by orchestrating an automated text-scrubbing pipeline using **Microsoft Presidio** to redact customer data prior to model invocation.
- **Eliminated LLM hallucinations in compliance workflows** to achieve **zero unauthorized generative text output** by implementing strict context-grounding constraints and delivering a **citation-backed retrieval engine** with exact page verification.
- **Transformed unstructured financial PDFs into deterministic business actions** across **three distinct triage categories** by designing structured data extraction layers using **LangChain and Pydantic** to automatically route, flag, or escalate applications.
- **Enforced zero-data-loss resiliency across data pipelines** during third-party API rate limits and failures by integrating **Amazon SQS Dead Letter Queues (DLQ)** and implementing **exponential backoff with jitter** retry logic.
- **Hardened system security infrastructure** to achieve absolute isolation of customer data by implementing least-privilege **AWS IAM** execution policies and enforcing **AWS KMS** encryption at rest and in transit.

California State University, Fullerton

Fullerton, CA

Graduate Software Developer

10/2024 - 06/2025

- **Achieved 40% helpdesk ticket deflection and 98% retrieval accuracy** by engineering a citation-backed student resource platform using **Python, FastAPI, ChatGPT Edu, and RAG**, indexing **10,000+ institutional documents**.
- **Reduced median response latency by 70%** and minimized redundant API calls across production endpoints by implementing a **Redis semantic caching layer** integrated with **REST APIs** and automated terminal agents.
- **Reduced LLM operational costs by routing 82% of low-complexity requests** to locally hosted, quantized open-source models through an intent-aware **model-routing service**, bypassing premium API calls for routine queries.
- **Processed 3,000+ weekly bookings with zero double-booking errors** across **15+ high-demand campus spaces** by developing a backend reservation engine using **Express and JavaScript** with conflict-detection logic.
- **Reduced peak-hour campus congestion by 40%** for a student population of **25,000+ users** by building a real-time occupancy platform using **React, Node.js, and WebSockets** to stream live capacity metrics across gyms, study rooms, and parking lots.

Accenture

Hyderabad, India

Software Engineer

08/2022 - 07/2024

- **Developed and maintained core REST endpoints and Kafka consumers** using **Java 17 and Spring Boot**, contributing directly to a **250ms inventory synchronization latency SLO** for a high-volume retail data platform.
- **Implemented robust exception handling and validation rules** within **Spring Integration and Spring Batch** pipelines, capturing and filtering out **95% of malformed transaction records** prior to production database ingestion.
- **Streamlined CI/CD pipelines to support a 35% boost in deployment velocity** by containerizing services using **Docker**, updating **GitHub Actions** deployment files, and modifying modular **Terraform** infrastructure configurations.
- **Built centralized observability workflows that cut incident triage time by 30%**, configuring custom **Splunk dashboards** and **AWS CloudWatch** alerts to flag microservice error spikes and database pool degradation early.
- **Owned feature ticket execution and integration testing** in an **Agile** environment, collaborating with senior engineers and stakeholders to ensure system stability and support high-concurrency peak traffic during **Black Friday** events.

Accenture

Hyderabad, India

Software Engineer Co-Op

01/2022 - 07/2022

- **Implemented 5+ backend REST API endpoints** using **Java and Spring Boot**, improving transaction processing and data validation across core enterprise services while applying secure request handling and relational **SQL** validation.
- **Resolved 8+ production-critical defects** by debugging **REST APIs**, optimizing **SQL validation scripts**, and troubleshooting backend integration issues across enterprise application workflows.
- **Contributed to technical documentation** by documenting API contracts, system workflows, and architectural decisions while participating in **Agile/Scrum sprint planning, daily stand-ups, code reviews, and retrospectives**.

Projects

Semantic Code Intelligence Platform | *Python, FAISS, BM25, Cross-Encoders, React, FastAPI*

GitHub Link

- **Designed a hybrid code-retrieval platform indexing 406K+ lines across 2,464 files** (38K+ semantic chunks) leveraging **FAISS** dense retrieval, **BM25** lexical search, and **Reciprocal Rank Fusion (RRF)**.
- **Achieved 1.24s end-to-end retrieval latency** using incremental indexing, Cross-Encoder reranking, and decoupled **FastAPI** microservices with exact source symbol citations.

Open-SWE | Open-Source Contributor | *Python, Docker, LLM Tool Calling, Sandbox Runtimes*

- **Engineered secure Docker sandbox execution environments** using LangGraph, scaling the autonomous software engineering runtime across **5 merged pull requests** to seamlessly support pluggable multi-provider LLM API adapters.
- **Expanded core capability primitives** by implementing curated filesystem, code-search, and version control tools (e.g., `grep`, `file_editor`), automating end-to-end pull request delivery workflows directly linked to GitHub webhook triggers.
- **Refined the loop verification sub-agent architecture**, optimizing the interaction loop between the Programmer graph and a dedicated Reviewer sub-agent to iteratively catch runtime errors and feed targeted logs back into the execution flow.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, HTML5, CSS3

Frameworks & Libraries Spring Boot, FastAPI, Node.js, Express, React.js, Next.js, Redux, Tailwind CSS

Distributed Systems & Databases: REST APIs, Microservices, GraphQL, gRPC, PostgreSQL, MySQL, Redis, Amazon RDS

AI & LLM Engineering: Retrieval-Augmented Generation (RAG), Vector Search (FAISS), Cross-Encoder Reranking, LangChain, AI Agents, Tool Calling

Cloud & DevOps: AWS (S3, Lambda, CloudWatch), Kubernetes, Docker, Terraform, CI/CD (GitHub Actions), Prometheus, Grafana, OpenTelemetry, Splunk

Observability & Testing: Prometheus, Grafana, OpenTelemetry, Splunk, JUnit, Mockito, Jest, TDD, Agile/Scrum